

Dilangi Thilakarathna

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Profile

Software Engineer with 3+ years of **Java**, **backend**, **mobile development** and **production AWS deployment expertise** across fintech, life sciences, and e-commerce. MSc Computing graduate (Distinction, 2024) with AWS Certified Cloud Practitioner certification and hands-on experience building and deploying secure microservices to cloud infrastructure. Proven ability to design RESTful APIs, optimize CI/CD pipelines, and deliver scalable solutions from development through production. Seeking a backend or cloud engineering role.

Key Skills

- **Technical:** Java, Spring Boot, REST APIs, Microservices, AWS (EC2, S3, RDS, VPC, Lambda, DynamoDB, CloudFront), Production Deployment, MySQL, Docker, CI/CD (Jenkins, Terraform, GitHub Actions), Unit Testing (JUnit, Mockito), Cloud-Native Architecture
- **Additional:** JavaScript, Node.js, Python, Terraform, Kubernetes, Kafka, Android, Kotlin, Firebase
- **Practices:** OOP, Design Patterns, TDD, Agile, Problem-solving, Code Review

Professional Experience

Software Engineer (Placement) | [IDBS, UK](#) | Sept 2023 – Aug 2024

- Optimized CI/CD pipelines using **Jenkins** and **Terraform**, reducing deployment time by 40%
- Improved Snowflake query performance by 50% through materialized view implementation for 5 critical dashboard queries
- Enhanced AWS infrastructure including DynamoDB operations, CloudFront security, and RDS database optimization
- Increased **unit test coverage** from 45% to 75% using **JUnit** and **Mockito**
- Contributed to AWS CloudFront security configuration
- Collaborated with cross-functional teams on Tableau dashboards and analytics pipelines

Software Engineer | [IFS R&D International, Sri Lanka](#) | Mar 2022 – Sept 2022

- Contributed to **ERP product** modules using proprietary frameworks
- Performed **database optimization** and involved in the full software lifecycle
- Employment ended due to overseas relocation for postgraduate studies.

Software Engineer | [Kapruka.com, Sri Lanka](https://www.kapruka.com) | Dec 2020 – Mar 2022

- Designed and built delivery management Android app from scratch as sole mobile developer
- Architected route optimization algorithm, reducing average delivery time by 25%
- Integrated Firebase Cloud Messaging for real-time notifications to 100+ drivers
- Deployed to Google Play Store

Associate Software Engineer | [Epic Lanka, Sri Lanka](https://www.epic.lanka) | Mar 2019 – Sept 2020

- Developed mobile apps for **fintech** services including QR payment systems
- Integrated front-end and back-end components for banking applications

Technical Projects

[Spring Boot Microservices API with Authentication](#) (2025)

- Designed and deployed production-grade RESTful API with JWT authentication to AWS
- Implemented secure user authentication system with endpoints using Spring Security and BCrypt
- Architected cloud-native deployment: AWS EC2 compute, RDS MySQL database, VPC security configuration
- Built comprehensive CRUD operations with proper error handling and input validation
- Unit test coverage using JUnit and Mockito for reliable production releases
- Live Demo: Available on request (deployed to AWS EC2 + RDS, currently offline to minimize costs)

[AI-Driven Energy Efficiency Optimization for Buildings](#) (Self-Learning project, 2024)

- Analyzed building energy consumption dataset (500+ records) using Python and scikit-learn
- Evaluated 5 regression models for heating/cooling load prediction, achieving $R^2 > 0.95$
- Identified polynomial regression as optimal model for energy efficiency estimation

[IoT Smart Door System](#) (MSc Project, 2023)

- Designed and implemented prototype of IoT-enabled smart door system using Arduino, Python, and Bluetooth communication
- Developed an Android application for real-time door status monitoring and alert notifications
- Documented system architecture in IEEE-format technical paper

Education

- **MSc. Computing** (Distinction, includes 12-months placement)
University of Northampton, UK | 2022 - 2024
- **BSc. Computer Science** (2:2)
University of Colombo School of Computing, Sri Lanka | 2016 - 2019

Certifications

- [AWS Certified Cloud Practitioner](#) (Oct 2025)